Free pythagorean theorem worksheet worksheet 29

Maths | 10th grade | Mixed Operations

1. If x = 5 and y = -3, calculate the value of 3x^2 - 2xy + y^2.

A) 48
B) 56
C) 64
D) 72

2. Solve for x: 4(x - 5) + 6 = 3x + 2.

A) x = 16
B) x = 14
C) x = 18
D) x = 20

3. Find the result of (2/3)^2 + (4/5)^2.

A) 0.789
B) 1.056
C) 0.874
D) 1.112

4. Evaluate the expression: (x^2 - y^2)/(x + y) where x = 8 and y = 3.

A) 11
B) 15
C) 5
D) 13

5. Determine the value of |2x - 5| + |x + 3| when x = -2.

A) 5
B) 7
C) 9
D) 11

6. If a and b are roots of x^2 - 5x + 6 = 0, calculate a^2 + b^2.

A) 25
B) 13
C) 10
D) 20

7. Simplify the expression: (3x^2y - 2xy^2)/(xy).

A) 3x - 2y
B) 3x + 2y
C) x^2 - 2y
D) 3x - 2y^2

8. Find the sum of the first 10 terms of the arithmetic sequence 3, 7, 11...

A) 210
B) 220

C) 230
D) 240

9. What is the remainder when 3x^3 - 2x^2 + 4x - 7 is divided by x - 2?

A) 3
B) 5
C) -1
D) 0

10. If log base 3 (x) = 4, what is x?

A) 64
B) 81
C) 72
D) 100

Answer Key

1. A
Substitute x = 5 and y = -3 into the expression: 3(5)^2 - 2(5)(-3) + (-3)^2 = 3(25) + 30 + 9 = 75 + 30 + 9 = 114.

2. A
Expand and simplify: 4x - 20 + 6 = 3x + 2 -> 4x - 14 = 3x + 2 -> x = 16.

3. C
(2/3)^2 = 4/9, (4/5)^2 = 16/25. Calculate: 4/9 + 16/25 = 100/225 + 144/225 = 244/225 = 0.874.

4. B
Factor the numerator: (x^2 - y^2) = (x - y)(x + y), cancels with denominator: (x - y) = 8 - 3 = 5.

5. D
Calculate: |2(-2) - 5| + |-2 + 3| = |-9| + |1| = 9 + 1 = 10.

6. C
a + b = 5 and ab = 6. a^2 + b^2 = (a + b)^2 - 2ab = 25 - 12 = 13.

7. A
Factor out xy from the numerator: xy(3x - 2y)/xy = 3x - 2y.

8. A
First term a = 3, common difference d = 4, n = 10. Sum = n/2 * (2a + (n-1)d) = 5 * (6 + 36) = 210.

9. C
Using the Remainder Theorem, substitute x = 2 into the polynomial: 3(2)^3 - 2(2)^2 + 4(2) - 7 = 24 - 8 + 8 - 7 = 17.

10. B
x = 3^4 = 81.